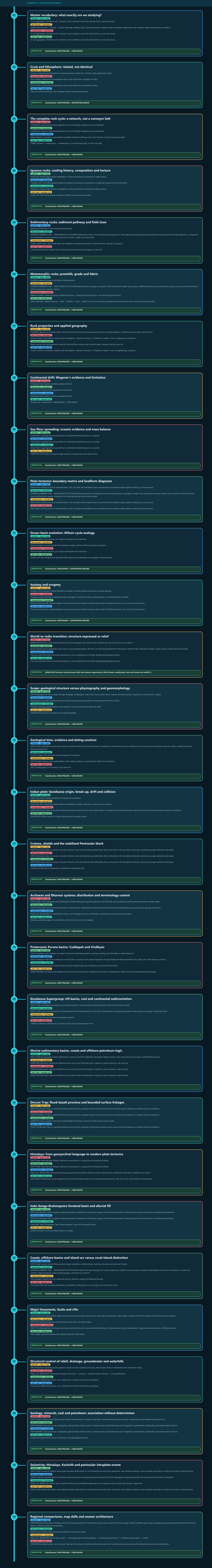

01

Master vocabulary: what exactly are we studying?

CONTEXT + EXACT CORE

A rock aggregate is therefore not “a mineral”, and a landform cannot be inferred from a rock name alone.

MECHANISM / ARGUMENT

[ANALYSIS] Structure → process → surface: lithology, bedding, joints, faults and tectonic setting create a template; weathering, rivers, glaciers, waves and human use modify it.

CONSEQUENCE / CONTRAST

A rock aggregate is therefore not “a mineral”, and a landform cannot be inferred from a rock name alone.

UPSC TRAP / ANSWER USE

A rock aggregate is therefore not “a mineral”, and a landform cannot be inferred from a rock name alone.

ANSWER LINE: **Classification: CORE PRELIMS + CORE MAINS**

02

Crust and lithosphere: related, not identical

CONTEXT + EXACT CORE

Crust is compositional; lithosphere is mechanical and includes the crust plus rigid uppermost mantle.

MECHANISM / ARGUMENT

The contrast, mechanism and exceptions earn more marks than a doubtful number.

CONSEQUENCE / CONTRAST

The contrast, mechanism and exceptions earn more marks than a doubtful number.

UPSC TRAP / ANSWER USE

They are useful mnemonics, not complete modern compositional layers.

ANSWER LINE: **Classification: CORE PRELIMS + SUPPORTING MAINS**

03

The complete rock cycle: a network, not a conveyor belt

CONTEXT + EXACT CORE

Lithification means turning loose sediment into rock through compaction and cementation.

MECHANISM / ARGUMENT

Lithification means turning loose sediment into rock through compaction and cementation.

CONSEQUENCE / CONTRAST

Metamorphism normally occurs without complete melting; if melting occurs, the material re-enters the igneous path.

UPSC TRAP / ANSWER USE

[TRAP] “Igneous → sedimentary → metamorphic” is one teaching path, not the only path.

ANSWER LINE: **Classification: CORE PRELIMS + CORE MAINS**

04

Igneous rocks: cooling history, composition and texture

CONTEXT + EXACT CORE

Porphyritic means large crystals embedded in a finer groundmass, showing two-stage cooling.

MECHANISM / ARGUMENT

“Primary rock” is an older genetic phrase for igneous rock because crystallisation creates the original mineral framework.

CONSEQUENCE / CONTRAST

Porphyritic means large crystals embedded in a finer groundmass, showing two-stage cooling.

UPSC TRAP / ANSWER USE

It does not mean every igneous exposure is oldest or economically primary.

ANSWER LINE: **Classification: CORE PRELIMS + CORE MAINS**

05

Sedimentary rocks: sediment pathway and field clues

CONTEXT + EXACT CORE

Bedding/stratification means depositional layering.

MECHANISM / ARGUMENT

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM Sedimentary rocks convert weathered, transported or chemically and biologically produced material into layered rock through deposition, compaction and cementation, while preserving clues such as fossils, ripples and mud cracks.

CONSEQUENCE / CONTRAST

Fossils, ripple marks, mud cracks and cross-bedding can reveal environment, current direction, drying or younging.

UPSC TRAP / ANSWER USE

Limestone has several origins; do not call every limestone exclusively organic or chemical.

ANSWER LINE: **Classification: CORE PRELIMS + CORE MAINS**

06

Metamorphic rocks: protolith, grade and fabric

CONTEXT + EXACT CORE

Protolith means the parent rock before metamorphism.

MECHANISM / ARGUMENT

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM Metamorphism changes a protolith in the solid state through heat, pressure, differential stress and reactive fluids, producing new minerals and fabrics without requiring complete melting.

CONSEQUENCE / CONTRAST

Agents are heat, confining pressure, differential stress—unequal directed pressure—and chemically active fluids.

UPSC TRAP / ANSWER USE

Exam-safe paths: shale/mudstone → slate → phyllite → schist → gneiss is a common increasing-grade teaching sequence, not a universal compulsory path.

ANSWER LINE: **Classification: CORE PRELIMS + CORE MAINS**

07

Rock properties and applied geography

CONTEXT + EXACT CORE

Crystalline igneous and metamorphic rocks commonly have low primary pore space but can store groundwater in weathered zones, joints and fractures.

MECHANISM / ARGUMENT

A dam, tunnel or foundation requires site investigation; “granite is strong” or “limestone is weak” is not an engineering conclusion.

CONSEQUENCE / CONTRAST

Basalt aquifers are heterogeneous: vesicular tops and flow contacts may transmit water; compact interiors may not.

UPSC TRAP / ANSWER USE

A dam, tunnel or foundation requires site investigation; “granite is strong” or “limestone is weak” is not an engineering conclusion.

ANSWER LINE: **Classification: CORE PRELIMS + CORE MAINS**

08

Continental drift: Wegener’s evidence and limitation

CONTEXT + EXACT CORE

Gondwana here means a southern palaeocontinent.

MECHANISM / ARGUMENT

Ocean-floor evidence later supplied the mechanism.

CONSEQUENCE / CONTRAST

Gondwana here means a southern palaeocontinent.

UPSC TRAP / ANSWER USE

Answer-use: Classification: CORE PRELIMS + CORE MAINS

ANSWER LINE: **Classification: CORE PRELIMS + CORE MAINS**

09

Sea-floor spreading: oceanic evidence and mass balance

CONTEXT + EXACT CORE

Ocean floor is geologically young relative to continental shields because it is recycled.

MECHANISM / ARGUMENT

Ocean floor is geologically young relative to continental shields because it is recycled.

CONSEQUENCE / CONTRAST

Ocean floor is geologically young relative to continental shields because it is recycled.

UPSC TRAP / ANSWER USE

[TRAP] The Earth does not expand; ridge creation is balanced by trench destruction.

ANSWER LINE: **Classification: CORE PRELIMS + CORE MAINS**

10

Plate tectonics: boundary matrix and landform diagnosis

CONTEXT + EXACT CORE

Plate-interior activity occurs at inherited faults, rifts, hot spots and intraplate stress concentrations; boundaries explain global clustering, not every event.

MECHANISM / ARGUMENT

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM Plate boundaries are diagnosed by motion and products: divergence creates crust, subduction destroys oceanic crust, continent collision thickens crust, and transform motion conserves crust while generating shallow earthquakes.

CONSEQUENCE / CONTRAST

Plate-interior activity occurs at inherited faults, rifts, hot spots and intraplate stress concentrations; boundaries explain global clustering, not every event.

UPSC TRAP / ANSWER USE

Plate-interior activity occurs at inherited faults, rifts, hot spots and intraplate stress concentrations; boundaries explain global clustering, not every event.

ANSWER LINE: **Classification: CORE PRELIMS + CORE MAINS**

11

Ocean-basin evolution: Wilson-cycle analogy

CONTEXT + EXACT CORE

Active margins carry trenches, arcs, major earthquakes and volcanism.

MECHANISM / ARGUMENT

Passive margins can accumulate thick sediment wedges and host offshore petroleum systems.

CONSEQUENCE / CONTRAST

Active margins carry trenches, arcs, major earthquakes and volcanism.

UPSC TRAP / ANSWER USE

A basin can stall, reorganise or avoid the ideal sequence; the examples are analogies, not timestamps.

ANSWER LINE: **Classification: CORE MAINS + SUPPORTING PRELIMS**

12

Isostasy and orogeny

CONTEXT + EXACT CORE

Isostasy means buoyant vertical equilibrium between crust/lithosphere and denser material beneath.

MECHANISM / ARGUMENT

Orogeny means mountain building through convergence, folding, thrusting, metamorphism, crustal thickening and uplift.

CONSEQUENCE / CONTRAST

Erosion removes mass, exposes deeper rocks and can produce isostatic response; high relief is therefore dynamic, not a one-time construction.

UPSC TRAP / ANSWER USE

Erosion removes mass, exposes deeper rocks and can produce isostatic response; high relief is therefore dynamic, not a one-time construction.

ANSWER LINE: **Classification: CORE MAINS + SUPPORTING PRELIMS**

13

World-to-India transition: structure expressed as relief

CONTEXT + EXACT CORE

[ANALYSIS] Structure controls broad relief and resource opportunity, while climate, weathering, rivers and human use modify it.

MECHANISM / ARGUMENT

At the highest school-geography scale, India’s six broad physiographic divisions are the Himalaya/Northern Mountains, Northern Plain, Peninsular Plateau, Indian Desert, Coastal Plains and Islands.

CONSEQUENCE / CONTRAST

They are used here only as structural expressions, not as substitutes for the later detailed physiography owners.

UPSC TRAP / ANSWER USE

They are used here only as structural expressions, not as substitutes for the later detailed physiography owners.

ANSWER LINE: **[ANALYSIS] Structure controls broad relief and resource opportunity, while climate, weathering, rivers and human use modify it.**

14

Scope: geological structure versus physiography and geomorphology

CONTEXT + EXACT CORE

[FACT] Geological structure means the age, lithology, stratigraphic order, folds, faults, joints, basins, cratons and plate-tectonic relations of rocks beneath a region.

MECHANISM / ARGUMENT

Physiography describes the broad surface divisions; geomorphology explains the processes and forms of relief.

CONSEQUENCE / CONTRAST

Therefore geology explains pattern and constraint, not every present landform by itself.

UPSC TRAP / ANSWER USE

[ANALYSIS] The three are linked but not interchangeable.

ANSWER LINE: **Classification: CORE PRELIMS + CORE MAINS**

15

Geological time, evidence and dating cautions

CONTEXT + EXACT CORE

[FACT] India's geological record spans ancient crystalline basement, Proterozoic basins, Gondwana continental successions, Mesozoic marine and volcanic episodes, Cenozoic collision-related strata and Quaternary alluvium.

MECHANISM / ARGUMENT

Correct: Usage varies by author and stratigraphic convention.

CONSEQUENCE / CONTRAST

[FACT] Relative dating uses superposition, cross-cutting relations, unconformities, fossils and correlation.

UPSC TRAP / ANSWER USE

[FACT] Stratigraphy is not merely a rock-name list.

ANSWER LINE: **Classification: CORE PRELIMS + CORE MAINS**

16

Indian plate: Gondwana origin, break-up, drift and collision

CONTEXT + EXACT CORE

[FACT] The Indian continental block formed part of Gondwana.

MECHANISM / ARGUMENT

[FACT] Sea-floor spreading creates oceanic lithosphere at ridges; subduction consumes it at trenches.

CONSEQUENCE / CONTRAST

[ANALYSIS] India is both an ancient continental fragment and part of an active plate system: this explains the contrast between old shield rocks and young Himalayan tectonic architecture.

UPSC TRAP / ANSWER USE

[ANALYSIS] Collision was not a single instant across the whole margin.

ANSWER LINE: **Classification: CORE PRELIMS + CORE MAINS**

17

Cratons, shields and the stabilised Peninsular block

CONTEXT + EXACT CORE

[FACT] Peninsular India is a mosaic of ancient cratonic nuclei and intervening mobile belts, later cut by faults, rifts and dykes and locally covered by younger sediments and basalt.

MECHANISM / ARGUMENT

[FACT] Peninsular India is a mosaic of ancient cratonic nuclei and intervening mobile belts, later cut by faults, rifts and dykes and locally covered by younger sediments and basalt.

CONSEQUENCE / CONTRAST

[FACT] Peninsular India is a mosaic of ancient cratonic nuclei and intervening mobile belts, later cut by faults, rifts and dykes and locally covered by younger sediments and basalt.

UPSC TRAP / ANSWER USE

It does not mean flat, homogeneous, immobile or earthquake-free.

ANSWER LINE: **Classification: CORE PRELIMS + CORE MAINS**

18

Archaean and Dharwar systems: distribution and terminology control

CONTEXT + EXACT CORE

[FACT] Older Indian geography texts distinguish fundamental gneiss-granite basement from Dharwar supracrustal and metamorphosed sediment-volcanic belts.

MECHANISM / ARGUMENT

[FACT] These belts contain schists, quartzites, iron formations, metavolcanics and associated intrusions; exact stratigraphic grouping varies by geological province.

CONSEQUENCE / CONTRAST

Mains use: [ANALYSIS] In a classification answer, give lithology, structure, distribution, significance and a terminology caution.

UPSC TRAP / ANSWER USE

Modern geological usage is more refined, so the terms must not be collapsed.

ANSWER LINE: **Classification: CORE PRELIMS + CORE MAINS**

19

Proterozoic Purana basins: Cuddapah and Vindhyan

CONTEXT + EXACT CORE

[FACT] Cuddapah and Vindhyan are major Proterozoic sedimentary basin successions resting unconformably on older basement.

MECHANISM / ARGUMENT

[FACT] Vindhyan rocks form an extensive central Indian succession from eastern Rajasthan through Madhya Pradesh toward the Son valley, with other basin occurrences.

CONSEQUENCE / CONTRAST

Older textbooks group them under Purana, but their detailed ages and correlations are convention-sensitive.

UPSC TRAP / ANSWER USE

[ANALYSIS] Basin thickness and repeated unconformities show episodic subsidence and tectonic reorganisation rather than one uninterrupted quiet deposit.

ANSWER LINE: **Classification: CORE PRELIMS + CORE MAINS**

20

Gondwana Supergroup: rift basins, coal and continental sedimentation

CONTEXT + EXACT CORE

[FACT] India's Gondwana successions accumulated mainly in fault-bounded continental basins within the Peninsular shield.

MECHANISM / ARGUMENT

[FACT] Lower parts of the succession include glacially influenced deposits in some basins, followed by coal-bearing sandstone-shale cycles and younger continental strata.

CONSEQUENCE / CONTRAST

They are distinct from Gondwana the palaeocontinent.

UPSC TRAP / ANSWER USE

[LIMIT] Gondwana coalfields are not simply surface strips along modern rivers.

ANSWER LINE: **Classification: CORE PRELIMS + CORE MAINS**

21

Marine sedimentary basins, coasts and offshore petroleum logic

CONTEXT + EXACT CORE

[FACT] India also preserves marine sedimentary records in Kachchh, Rajasthan, Himalayan Tethyan sectors, east-coast basins and younger coastal/offshore basins.

MECHANISM / ARGUMENT

[ANALYSIS] Petroleum requires an effective source rock, burial and maturation, migration, porous reservoir, seal and trap.

CONSEQUENCE / CONTRAST

[ANALYSIS] Petroleum requires an effective source rock, burial and maturation, migration, porous reservoir, seal and trap.

UPSC TRAP / ANSWER USE

[ANALYSIS] Petroleum requires an effective source rock, burial and maturation, migration, porous reservoir, seal and trap.

ANSWER LINE: **Classification: CORE PRELIMS + CORE MAINS**

22

Deccan Trap: flood-basalt province and bounded surface linkages

CONTEXT + EXACT CORE

[FACT] The Deccan Trap is a vast flood-basalt province built by repeated, mainly fissure-fed lava flows around the end-Cretaceous to earliest Cenozoic transition.

MECHANISM / ARGUMENT

[FACT] The Deccan Trap is a vast flood-basalt province built by repeated, mainly fissure-fed lava flows around the end-Cretaceous to earliest Cenozoic transition.

CONSEQUENCE / CONTRAST

[LIMIT] The cause, duration and environmental effects of Deccan volcanism remain active research fields.

UPSC TRAP / ANSWER USE

[FACT] The Deccan Trap is a vast flood-basalt province built by repeated, mainly fissure-fed lava flows around the end-Cretaceous to earliest Cenozoic transition.

ANSWER LINE: **Classification: CORE PRELIMS + CORE MAINS**

23

Himalaya: from geosynclinal language to modern plate tectonics

CONTEXT + EXACT CORE

[FACT] Older texts describe Tethyan sediment accumulation in a geosyncline followed by folding.

MECHANISM / ARGUMENT

[FACT] Older texts describe Tethyan sediment accumulation in a geosyncline followed by folding.

CONSEQUENCE / CONTRAST

Mains use: [ANALYSIS] Contrast the historical geosynclinal description with the modern plate-tectonic mechanism; then draw a labelled cross-section.

UPSC TRAP / ANSWER USE

[ANALYSIS] 'Geosyncline' can be retained as a historical descriptive term for a long subsiding sedimentary belt, but not as a self-sufficient causal theory.

ANSWER LINE: **Classification: CORE PRELIMS + CORE MAINS**

24

Indo-Ganga-Brahmaputra foreland basin and alluvial fill

CONTEXT + EXACT CORE

[FACT] The Northern Plain occupies a foreland depression between the Himalayan orogen and Peninsular basement, filled by large volumes of Cenozoic and Quaternary sediment.

MECHANISM / ARGUMENT

[FACT] The Northern Plain occupies a foreland depression between the Himalayan orogen and Peninsular basement, filled by large volumes of Cenozoic and Quaternary sediment.

CONSEQUENCE / CONTRAST

Cross-link: Topic 05 drainage; Topic 04 groundwater; Topic 03 earthquake effects.

UPSC TRAP / ANSWER USE

[LIMIT] The plain is not one uniform layer or one age.

ANSWER LINE: **Classification: CORE PRELIMS + CORE MAINS**

25

Coasts, offshore basins and island-arc versus coral-island distinction

CONTEXT + EXACT CORE

[FACT] India's coasts record rifting, passive-margin subsidence, sedimentation, faulting, volcanism and sea-level change.

MECHANISM / ARGUMENT

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM Andaman-Nicobar belongs to an active subduction-related arc-forearc system, whereas Lakshadweep consists of coral atolls on a submarine volcanic-ridge foundation; Indian island geology is therefore not uniform.

CONSEQUENCE / CONTRAST

The emergent islands are coral, unlike the tectonic island-arc setting of Andaman-Nicobar.

UPSC TRAP / ANSWER USE

The island chain also includes sedimentary and uplifted components; it is not simply a row of volcanic cones.

ANSWER LINE: **Classification: CORE PRELIMS + CORE MAINS**

26

Major lineaments, faults and rifts

CONTEXT + EXACT CORE

[FACT] Peninsular India contains major inherited structural corridors including the Narmada-Son lineament, Tapi trough, Cambay rift and the Godavari and Mahanadi basin systems.

MECHANISM / ARGUMENT

[FACT] Narmada and Tapi flow west through structurally controlled troughs.

CONSEQUENCE / CONTRAST

[FACT] The Narmada-Son structural zone crosses central India and is associated with faulting, rift-valley segments, basin development, magmatism and seismicity in different sectors.

UPSC TRAP / ANSWER USE

Their valleys cannot be explained by regional eastward slope alone.

ANSWER LINE: **Classification: CORE PRELIMS + CORE MAINS**

27

Structural control of relief, drainage, groundwater and waterfalls

CONTEXT + EXACT CORE

[FACT] Resistant quartzite, granite, gneiss or basalt can form uplands and scarps, while weaker shale or weathered zones erode into valleys.

MECHANISM / ARGUMENT

Mains use: [ANALYSIS] Frame every example as structure -> process -> landform/water outcome -> local qualification.

CONSEQUENCE / CONTRAST

The result is differential erosion, not a simple hard-rock/soft-rock binary everywhere.

UPSC TRAP / ANSWER USE

The result is differential erosion, not a simple hard-rock/soft-rock binary everywhere.

ANSWER LINE: **Classification: CORE PRELIMS + CORE MAINS**

28

Geology, minerals, coal and petroleum: association without determinism

CONTEXT + EXACT CORE

[FACT] India's resource map reflects ancient shield mineralisation, Gondwana coal basins, sedimentary petroleum systems, coastal placers and weathering-derived ores.

MECHANISM / ARGUMENT

[FACT] Metallic ores such as iron, manganese, gold and base metals cluster in several ancient shield belts because of magmatic, hydrothermal, sedimentary and metamorphic histories.

CONSEQUENCE / CONTRAST

[FACT] Metallic ores such as iron, manganese, gold and base metals cluster in several ancient shield belts because of magmatic, hydrothermal, sedimentary and metamorphic histories.

UPSC TRAP / ANSWER USE

It does not imply that all critical minerals lie in one geological province.

ANSWER LINE: **Classification: CORE PRELIMS + CORE MAINS**

29

Seismicity: Himalaya, Kachchh and peninsular intraplate events

CONTEXT + EXACT CORE

[FACT] India's seismicity reflects active plate-boundary deformation in the Himalaya and north-east, subduction near Andaman-Nicobar, and intraplate reactivation in Kachchh and parts of the Peninsula.

MECHANISM / ARGUMENT

[LIMIT] Disaster response doctrine, building-code enforcement and preparedness are owned by Disaster Management; geology provides the source and site-condition framework.

CONSEQUENCE / CONTRAST

Mains use: [ANALYSIS] Explain source, path/site effect and vulnerability separately; do not equate tectonic activity with disaster magnitude.

UPSC TRAP / ANSWER USE

[FACT] India's seismicity reflects active plate-boundary deformation in the Himalaya and north-east, subduction near Andaman-Nicobar, and intraplate reactivation in Kachchh and parts of the Peninsula.

ANSWER LINE: **Classification: CORE PRELIMS + CORE MAINS**

30

Regional comparisons, map skills and answer architecture

CONTEXT + EXACT CORE

[ANALYSIS] Peninsula versus Himalaya: ancient composite shield with inherited structures versus active fold-thrust orogen; relative stability versus high convergence; exposed basement/mineral belts versus tectonic relief and sediment production.

MECHANISM / ARGUMENT

[FACT] Cross-sections show mechanism better than decorative maps.

CONSEQUENCE / CONTRAST

Mains use: [ANALYSIS] Use the spine: claim -> named geological evidence/region -> mechanism/significance -> limitation/qualification -> verdict.

UPSC TRAP / ANSWER USE

[ANALYSIS] Gondwana basin versus Deccan Trap: fault-bounded continental sedimentary basin with coal measures versus extrusive flood-basalt province with stacked flows.

ANSWER LINE: **Classification: CORE PRELIMS + CORE MAINS**